Brain states representing Caring and Uncaring Others 2

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Intro: background and Motivation

- Aim: Use neuroscience to overcome the limitations of psychotherapy.
- Using evidence about others being caring or not has huge consequences for therapeutic relationship.

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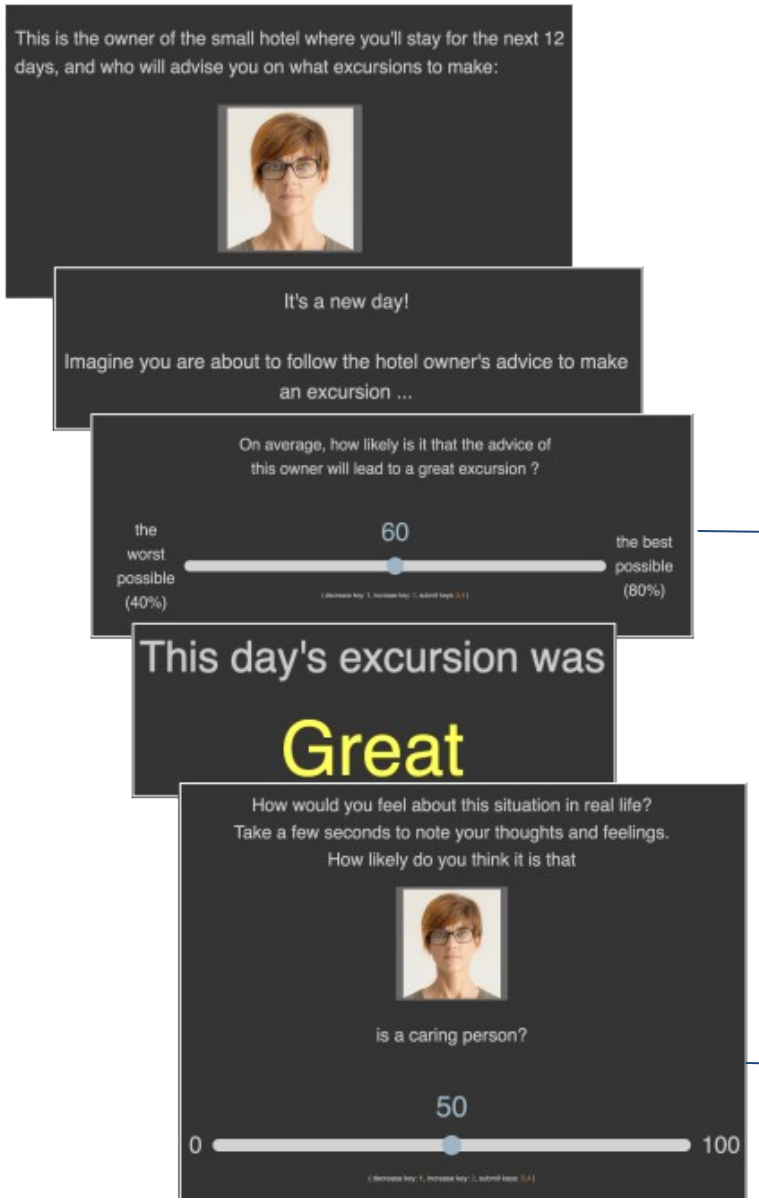
- Aim: Use neuroscience to overcome the limitations of psychotherapy.
- Using evidence about others being caring or not has huge consequences for therapeutic relationship.
- Is Valence Partitioned Temporal Difference learning used by the brain to decide whether others are caring or not ?
- Evidence for VPTD: Sands et al, 2024 (electrochemistry), our pilot work, unpublished (behavioural, two ‘social’ tasks).

Methods: Caring Attributions Task



State 1

State 2

Advice Estimate

State 3

Care inference

State 4

Care attribution

Hypotheses

- H1: The EEG signature of prediction errors accumulating Caring vs. Uncaring representations is Valence-partitioned (differentially timed VPPEs)

- H2: Caring vs. Uncaring others attract distinct ‘social brain’ representations.

Hypotheses and Peer Feedback

- H1: The EEG signature of prediction errors accumulating Caring vs. Uncaring representations is Valence-partitioned (differentially timed VPPEs)

- H2: Caring vs. Uncaring others attract distinct ‘social brain’ representations.

- PF1: There is no control condition separate from Caring-Uncaring! Would the mechanisms identified here be specific to Caring-Uncaring judgements, or reflect other dimensions, covarying with this ?

- PF2: What added value does imaging bring ?

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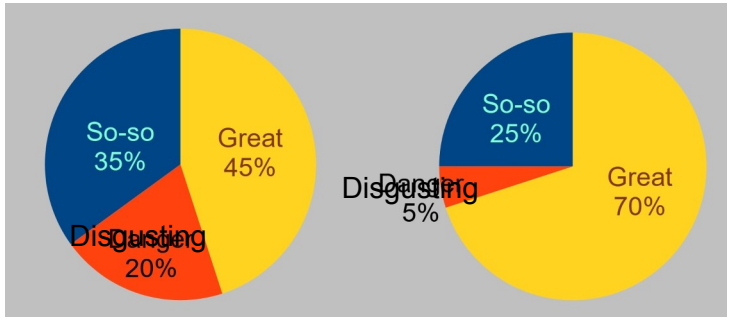
- You are an apple tree enthusiast staying in a series of small hotels. Every day, you are allowed to taste a complimentary apple from their apple tree.
- You are interested to guess whether the apple tree is well nourished or not.
- Whether the owner is caring or not has no impact at all on what kind of apple you pick.
- Well nourished trees provide 70% Great tasting apples, 25% So-so, and only 5% disgusting ones …



Added value of imaging

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As if the activity across all EEG sensors initially doesn't know whether to paint

the uncaring Queen of the Night,
or her caring daughter Pamina,

but the inferred face gradually emerges

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Classical interaction contrast

- `[caring - uncaring] - [well_nourished - malnourished]`

- Modelled VPPE ~ raw pre-processed voltage, states S1 to S4

Advantages and disadvantages

Advantages:

- Matched low-level features, valence, approximately matched cover story
- Within - participant comparison gives greater statistical power
- Financially more possible, two data blocks per recruited participant and EEG session.

Advantages and disadvantages

Disadvantages:

- Cognitive cross-contamination - ppl will either ‘anthropomorphise’ or ‘treat as a puzzle’, reducing control contrast

- Twice as long task - reduction in motivation and data quality due to fatigue and boredom.

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Mitigation:

- Non - interleaved blocks with break in between
- In N <16 **initial participants**, assess the quality of data of within-participant design.
- If signs of fatigue etc. detected, test **later participants** in an across-participant design, and **discard (or statistically discount)** the second sub-task of the initial participants in the analysis.

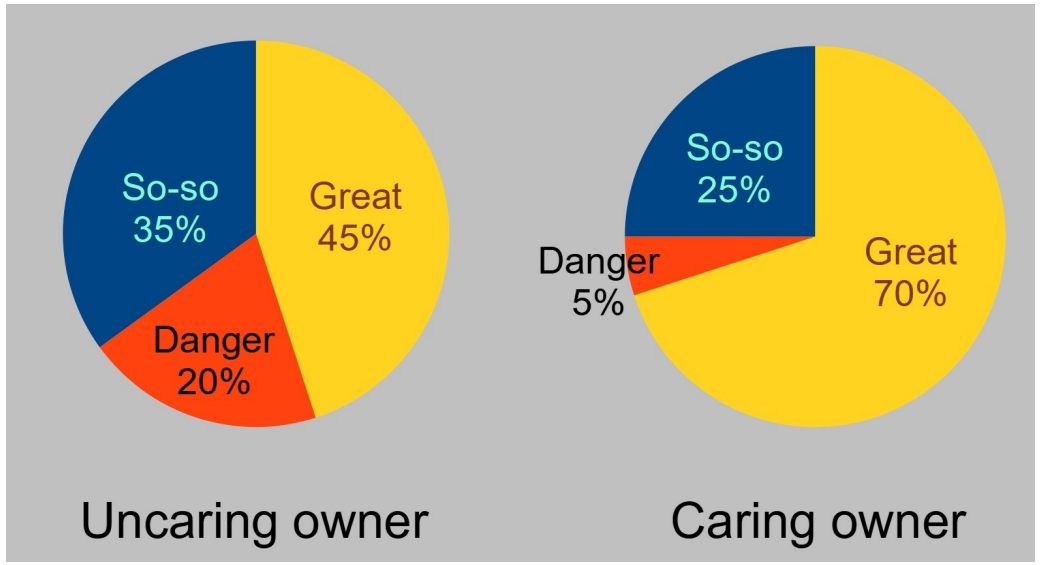
Discussion

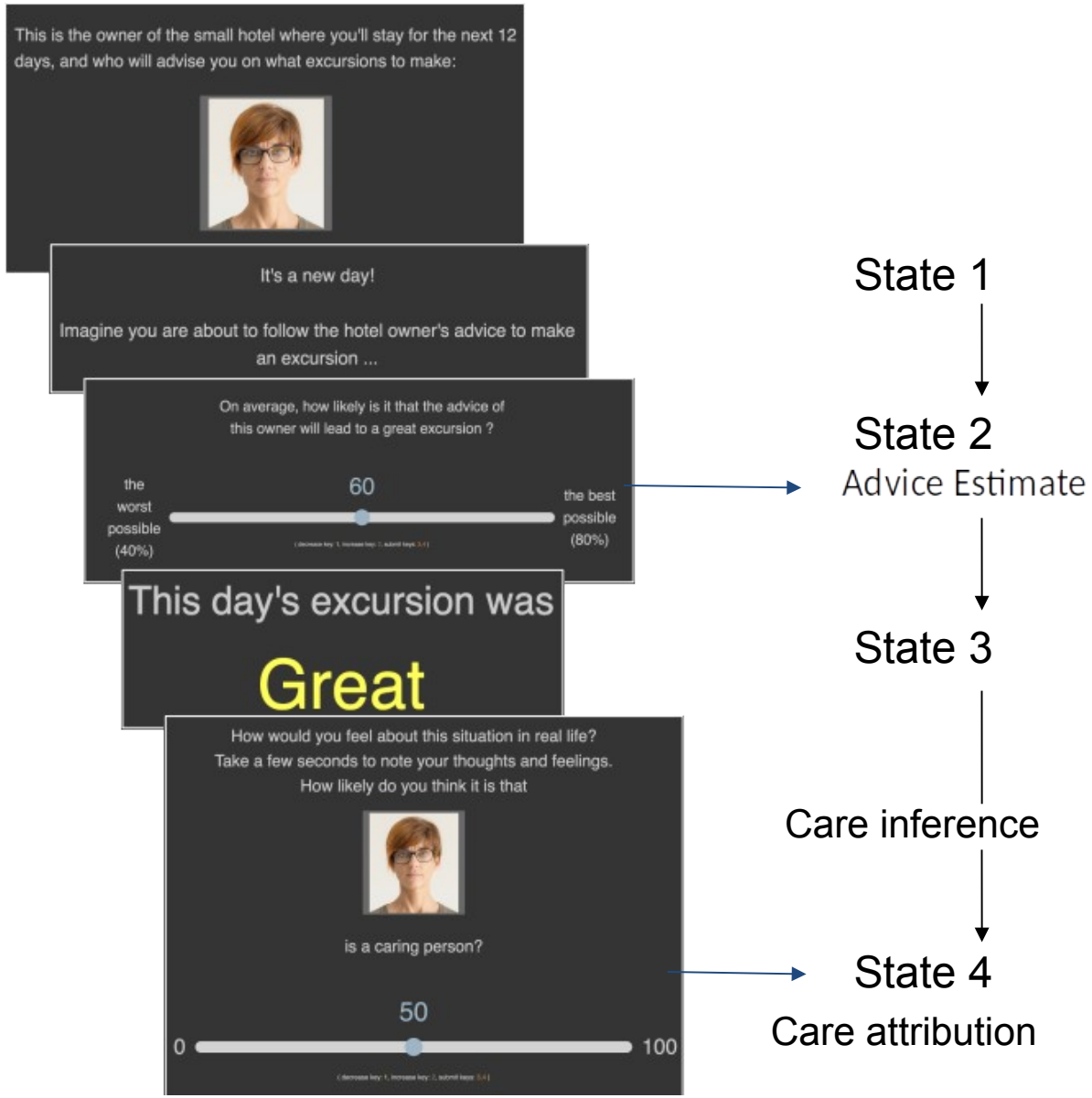
Methods: Caring Attributions Task

Imagine you are holidaying in faraway towns

The hotel owner gives you advice about where to go for your outing

Some hotel owners are caring, others are uncaring.

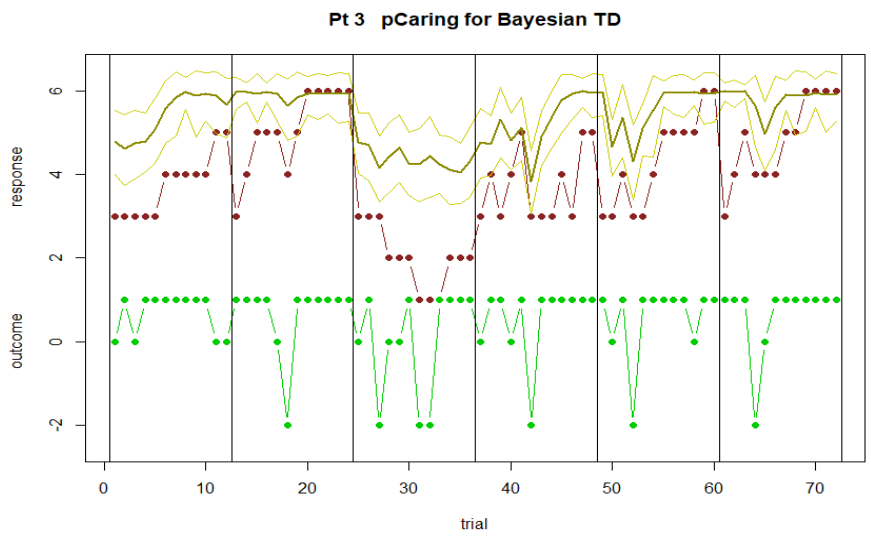




Methods: Analysis

Behavioural

- Associative Temporal-difference Value-partitioned modelling
- Bayesian – Temporal-difference hybrid Value-Partitioned modelling (novel)
- Prediction error parametric regressors for each of four states per trial



EEG

- Following well established pre-processing (Vladimir Litvak).
- GLM analysis of weighed prediction error estimates as parametric modulator for all four task states (Sands et al, 2024)
- Regression of preprocessed voltage vs. scalp sensors X time interval of interest
- Simple contrast of modeled Uncaring – Caring value over pre-action interval of ‘Care inference’

Logistics - cohort

- How many scanning hours are required per participant?
  - Standard 64 electrode EEG setup x 2 hours / participant
- Are there any cohort considerations?
  - Sensitive discussion of equality and inclusion with respect to culturally important issues e.g. hair.